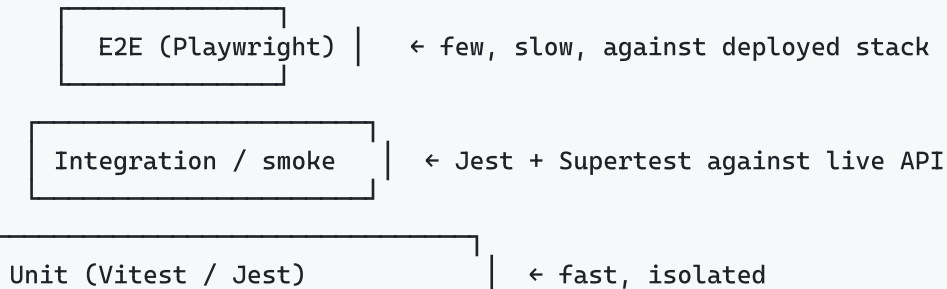


10 — Testing & QA

1. Test pyramid



Layer	Tool	Location	Status
Frontend unit/component	Vitest 3 + RTL + jsdom	frontend-react/src/**/*.__tests__ + frontend-react/vitest.config.ts	9 specs / passing
Backend smoke (live)	Jest 29 + Supertest	backend/test/smoke/	7 specs / passing
Backend e2e (legacy)	Jest 29	backend/test/	reserved
End-to-end	Playwright + Chromium	e2e/smoke/ + workspace playwright.config.ts	6 specs / passing

Last full run: 128/128 green (113 backend + 9 frontend + 6 E2E).

2. Frontend tests

```
cd frontend-react
npm install
npm run test          # vitest run
npm run test:watch
```

- Setup: [frontend-react/src/test/setup.ts](#) configures jsdom + RTL globals.
- Notable specs:
 - [VariantSelector.test.tsx](#) — verifies axes are hidden when only one option exists; sibling navigation works when multiple variants exist.
 - [cart.test.ts](#) — store reducer logic, totals, promo, loyalty.

3. Backend smoke tests

The smoke suite hits the **deployed** backend with a single shared QA customer to stay inside auth rate limits.

```
cd backend
$env:SMOKE_API_URL = 'https://backend-qlyb5greec2io.whiteriver-ae956411.eastus2.azurec
$env:SMOKE_ADMIN_EMAIL = 'admin@solo-ecommerce.com'
$env:SMOKE_ADMIN_PASSWORD = '<secret>'
npm run test:smoke
```

- Config: `backend/test/smoke/jest-smoke.json` → `maxWorkers: 1`.
- Global setup creates one customer `qa-smoke-{runId}@solo-qa.local`.
- Global teardown deletes the customer via `DELETE /api/admin/customers/:id` using an admin token.
- Specs cover: auth login/refresh, products list/detail, cart add/update, favorites add/remove, orders create (COD path), returns request, health.

Cleanup utility

If a run is interrupted, sweep stragglers with:

```
node backend/scripts/cleanup-smoke-data.js
```

It logs in as admin, lists customers matching `qa-smoke-*@solo-qa.local`, and deletes them with retry/back-off (60 retries, 30–60 s) — used after deploys to confirm `qa-smoke matches: 0`.

4. End-to-end (Playwright)

```
npx playwright install chromium      # one-time
npx playwright test e2e/smoke
```

- Config: `playwright.config.ts` (workspace root). `baseURL` points at the live Static Web App.
- Specs: home loads, navigation renders, product list paginates, product detail shows variants, cart add flow, header search.

5. Rate-limit guidance

`/api/auth/*` is hard-limited to **5 requests / 15 min / IP**. Tests must:

- Reuse a single login per run (smoke does this via `globalSetup`).
- Avoid registering a new user per spec.
- When iterating locally, wait at least 15 min after hitting the cap or switch IP.

6. Coverage matrix

Capability	Unit	Smoke	E2E
Login / refresh / logout	–	✓	–
Product list	–	✓	✓
Product detail incl. variants	✓ (frontend)	✓	✓
Cart CRUD	✓ (frontend store)	✓	✓
Favorites	–	✓	–
Order create (COD)	–	✓	–
Stripe checkout	–	–	manual
Returns request	–	✓	–
Admin RBAC	–	✓ (admin delete customer)	–
Health	–	✓	–

7. Manual QA checklists (lightweight)

Pre-release smoke (10 min) by a human:

1. Home loads; banners render; navigation tree opens.
2. Search returns results; product detail shows price + VAT line.
3. Add 2 products to cart; apply a promo code; remove one item.
4. Login as known customer; cart persists.
5. Checkout with Stripe test card `4242 4242 4242 4242`; order confirmation page; account → orders shows the new order; PDF invoice downloads.
6. Admin → orders → mark new order as `PROCESSING`; status history updated.
7. Logout → ensure `/account/*` routes redirect.

8. CI guidance

- Fast feedback: run **frontend unit** and **backend unit/lint** on every PR.
- Smoke: gated to runs against `staging` or post-deploy on `main`, serialised (`maxWorkers: 1`) to respect throttling.
- E2E: post-deploy on `main` against the deployed SWA.
- Always run `cleanup-smoke-data.js` as a final stage to avoid residue.

9. Quality gates before merge

-  Lint passes (`eslint`)
-  Type-check passes (`tsc --noEmit` / `nest build`)
-  Affected unit tests pass
-  No new `npm audit` HIGH/CRITICAL
-  Schema migrations reviewed and named descriptively
-  Public API change documented in [06 API Reference](#)